

13(2)

$$LHS = \frac{1}{z} - \frac{1}{z^2} \quad |z-1| > 1$$

$$\frac{1}{z} = \frac{1}{1+(z-1)} = \sum_{n=0}^{+\infty} (-1)^n (z-1)^n$$

$$\frac{1}{z^2} = -\left(\frac{1}{z}\right)' = -\sum_{n=1}^{+\infty} (-1)^{n-1} (-n) (z-1)^{n-1}$$

$$\therefore LHS = \sum_{n=1}^{+\infty} (-1)^n n (z-1)^{n-1}$$

13(4)

$$LHS = \sum_{n=0}^{+\infty} \frac{(-\frac{1}{z})^n}{n!} = \sum_{n=0}^{+\infty} \frac{(-1)^n}{n!} \frac{1}{z^n}$$

13(6) $-4 < z < 0$

$$\frac{1}{z} = -\frac{1}{z-(z+2)} = -\frac{1}{z} \sum_{n=0}^{+\infty} \left(\frac{z+2}{z}\right)^n = -\sum_{n=0}^{+\infty} \frac{(z+2)^n}{z^{n+1}}$$

$$LHS = -\sum_{n=0}^{+\infty} \frac{(z+2)^n}{z^{n+1}}$$

14. $\frac{1}{z-k} = \sum_{n=0}^{+\infty} \frac{k^n}{z^{n+1}}$

令 $z = e^{i\theta}$, $RHS = \sum_{n=0}^{+\infty} \frac{k^n}{e^{i\theta(n+1)}} = \sum_{n=0}^{+\infty} k^n [\cos(n+1)\theta - i \sin(n+1)\theta]$

$$LHS = \frac{1}{e^{i\theta} - k} = \frac{1}{(\cos\theta - k) + i \sin\theta} = \frac{\cos\theta - k - i \sin\theta}{1 - 2k \cos\theta + k^2}$$

$\therefore$ 得证